

Transformative and Subsistence Entrepreneurs: Origins and Impacts on Economic Growth

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Motivation & Entrepreneur Taxonomy

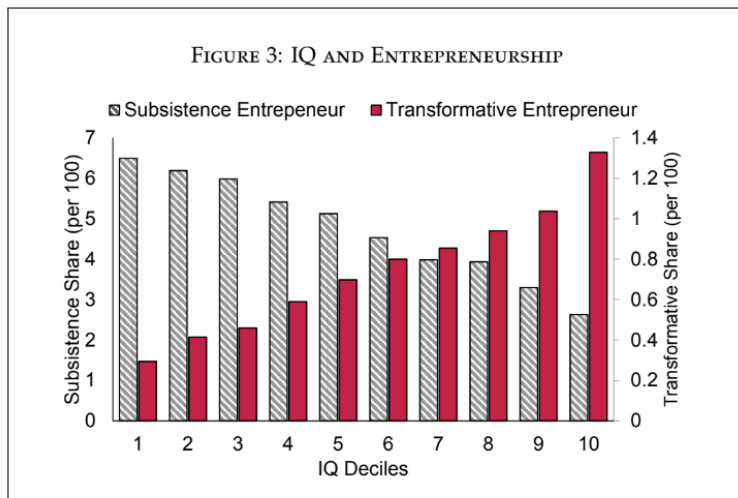
- **Heterogeneity in Entrepreneurship:** Most startups do not innovate or scale up (Hurst & Pugsley, 2011).
- **Novel Ex-Ante Classification (by intent, not ex-post success):**
 - **Transformative Entrepreneurs:** Hire R&D workers → aim to innovate, scale, and drive aggregate growth.
 - **Subsistence Entrepreneurs:** Do NOT hire R&D workers → focus on day-to-day operations with flat growth.

Empirical Regularities (Danish Microdata)

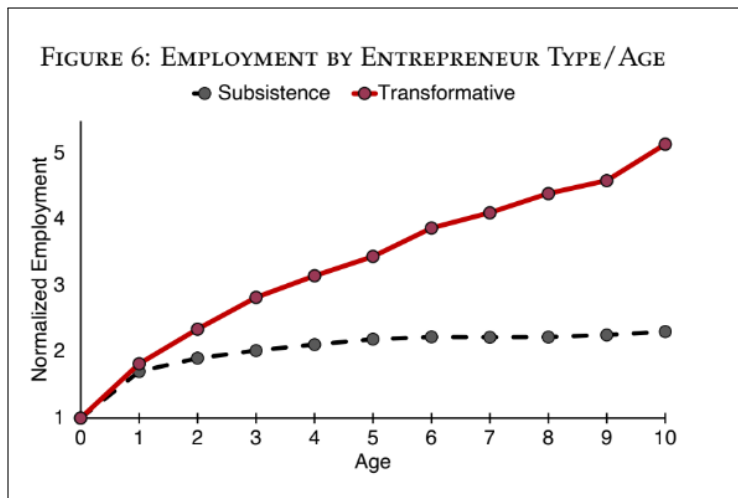
Key Facts from Danish Administrative Data

- 1 Aggregate Selection:** Overall entrepreneurship is *negatively* selected on IQ and education.
- 2 Family Background:** Parental entrepreneurship is the strongest predictor of founding a firm.
- 3 R&D Workers:** Strongly *positively* selected on IQ and higher education.
- 4 Transformative Founders:** Unlike aggregate founders, they resemble R&D workers—high IQ and college educated.
- 5 Firm Growth:** Transformative firms start larger, grow $> 9\%$ faster per year, and generate 34% of employment at age 10 (vs. 13% at founding).

Empirical Evidence: IQ Sorting by Entrepreneur Type



Empirical Evidence: Life-Cycle Employment Growth



Structural Model: Environment & Heterogeneity

- **Demographics:** Continuous time; new agents born at rate m , death rate ψ .
- **Initial Tri-variate Heterogeneity** drawn from $\Omega(z, y, f)$:
 - Worker ability / talent: $z \sim \text{Lognormal}$
 - Parental resources / wealth: $y \sim \text{Lognormal}$
 - Parental entrepreneurship background: $f \in \{0, 1\}$
- **Three Career Paths:**
 - 1 Production Worker ($s = 0$)
 - 2 R&D Worker ($s = 1$)
 - 3 Entrepreneur ($\theta \in \{0, \bar{\theta}\}$)

Dimensions of Heterogeneity

- **Individual Level** $\Omega(z, y, f)$:
 - Talent $z \implies$ Productivity α_s & transformative draw probability
 - Income $y \implies$ Credit constraint for education ($y \geq c_s$)
 - Parent $f \in \{0, 1\} \implies$ Idea arrival rate $\lambda_{f,s}$
- **Entrepreneur Level** $\theta \in \{0, \bar{\theta}\}$:
 - Transformative ($\bar{\theta}$): Hires R&D labor, innovates ($X > 0$)
 - Subsistence (0): Production labor only, no R&D ($X = 0$)
- **Firm Level** $n \geq 1$: Number of active product lines

Model: Career Choice & Entrepreneurial Entry

■ Education Choice ($s \in \{0, 1\}$):

- Credit friction: $y < c_s \implies s = 0$ (Production worker).
- Unconstrained ($y \geq c_s$): Logit choice for R&D worker ($s = 1$) based on $W_1 - W_0$.

■ Entrepreneurial Opportunity & Type Draw ($\theta \in \{0, \bar{\theta}\}$):

- Arrival rate $\lambda_{f,s}$ boosted by education ($s = 1$) and parent founder ($f = 1$).
- Transformative type probability $\Pi(\bar{\theta} \mid z, s)$ rises with ability z and schooling s .
- Entry condition: $V_e(\theta) - c_e > W_s$.

Model: Firm Innovation (Klette-Kortum)

■ Innovation Technology (n product lines):

$$X(n, \theta) = \theta \cdot l_{rd}^{\sigma} \cdot n^{\eta} \implies C(X) = \omega_1 \left(\frac{X}{\theta n^{\eta}} \right)^{\frac{1}{\sigma}}$$

■ Firm Heterogeneity in Production:

- **Transformative** ($\theta = \bar{\theta}$): Hires R&D labor (l_{rd}), innovates ($X > 0$), acquires lines.
- **Subsistence** ($\theta = 0$): Cannot do R&D ($X = 0$), grows only via exogenous rate v .

Model: Firm Value Function $V(n, \theta)$

Dynamic Bellman Equation

$$\begin{aligned}\rho V(n, \theta) = & \pi n + \tau n [V(n-1, \theta) - V(n, \theta)] \\ & + \max_X \left\{ X [V(n+1, \theta) - V(n, \theta)] - \omega_1 \left(\frac{X}{\theta n^\eta} \right)^{\frac{1}{\sigma}} \right\} \\ & + \nu n [V(n+1, \theta) - V(n, \theta)] + \psi [0 - V(n, \theta)]\end{aligned}$$

- πn : Static monopoly profits across n lines.
- τ : Creative destruction rate (losing a line to an innovator/entrant).
- ψ : Exogenous exit due to entrepreneur death.

General Equilibrium & Aggregate Growth

■ R&D Labor Market Clearing (Demand = Supply):

$$\sum_{\theta} \sum_n \left(\frac{X(n, \theta)}{\theta n^{\eta}} \right)^{\frac{1}{\sigma}} \Psi(n, \theta) = \sum_f \iint \Phi_1(z, y, f) z^{\alpha_1} dz dy$$

■ Aggregate Innovation Rate (τ_g):

$$\tau_g = \sum_{\theta} s_{\theta}^I e_{\theta} + \sum_{\theta} \sum_n X(n, \theta) \Psi(n, \theta)$$

■ Balanced Growth Path (BGP) Rate:

$$g = \ln(\gamma) \tau_g \quad (\text{where } \gamma \text{ is the innovation step size})$$

Counterfactual 1: Eliminating Financial Barriers

Experiment: Remove credit constraints ($y \geq c_s$ for all) $\implies$ education depends only on ability and preferences.

Variable	Baseline	No Frictions	% Change
Growth Rate (g)	1.00%	1.11%	+11.08%
R&D Workers	1.58%	1.82%	+15.24%
Transformative Entr.	0.50%	0.54%	+7.49%
Subsistence Entr.	1.59%	1.53%	-3.60%

Counterfactual 1: General Equilibrium Decomposition

Supply vs. Demand Decomposition

- **Supply-Only (Partial Equilibrium):** Fixing firm demand yields only a **+6.6%** growth increase.
- **Full General Equilibrium Effect:** Allowing both supply and demand to adjust yields a **+11.1%** growth increase.
- **Key Takeaway:** The demand side (transformative entrepreneurs absorbing inventors) accounts for **40% of the aggregate growth gain!**

Counterfactual 2: Comparing Policy Instruments

- **Setting:** Evaluate alternative policy tools using an identical budget of **0.05% of GDP**.

% Change wrt Baseline	Uniform Startup	Incumbent R&D	Innovative Startup	Education Subsidy
Growth Rate (g)	+0.11%	+2.31%	+4.38%	+16.14%
R&D Workers	-0.02%	+2.86%	+4.04%	+22.26%
Transformative Entr.	+0.50%	+1.49%	+6.69%	+10.84%
Subsistence Entr.	+0.76%	-0.74%	-1.45%	-5.27%
Production Workers	-0.01%	-0.04%	-0.08%	-0.33%

Policy Comparison: Key Insights

- **Uniform Subsidies (+0.11% growth):** Mainly attract subsistence founders who do not innovate or hire R&D labor.
- **Incumbent R&D Subsidies (+2.31% growth):** Boost research in existing firms but hit diminishing returns ($\sigma, \eta < 1$) and fail to expand the talent pool.
- **Innovative Startup Subsidies (+4.38% growth):** Effectively target growth-oriented entry and increase R&D demand.
- **Education Subsidies (+16.14% growth):** Tackles the root friction, expanding **both** R&D supply and transformative firm creation.

Optimal Policy Mix Across Budget Levels

Policy Pecking Order

- **Low Budgets ($\leq 0.50\%$ of GDP):** Allocate funding entirely to **Education Subsidies** and **Innovative Startup Subsidies**.
- **High Budgets ($> 0.50\%$ of GDP):** Introduce **Incumbent R&D Subsidies** once the underlying talent pipeline is sufficiently enlarged.
- **Core Message:** Subsidizing R&D without expanding the human capital pipeline simply inflates wages rather than driving innovation.

Main Conclusions

- 1 Symbiotic Relationship:** Long-run growth relies on the interplay between R&D talent (supply) and transformative entrepreneurs (demand).
- 2 Educational Bottlenecks:** Financial frictions in education create severe talent misallocation across both career paths.
- 3 Targeted Policy:** Education subsidies yield the highest leverage for innovation policy by strengthening both sides of the market.